CRYSTALLOGRAPHY & X-RAY DIFFRACTION — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: March 29, 2026

Diagram-based questions on Crystallography & X-ray Diffraction

Question 1

10.0 points

Examine the diagram below illustrating the geometry of Bragg's law, where an incident X-ray beam strikes a crystal plane and the diffracted beam is observed. Identify the angle of incidence, the spacing between planes, and the conditions for constructive interference according to the equation $n\lambda = 2d\sin\theta$. Explain how each component is represented in the diagram.

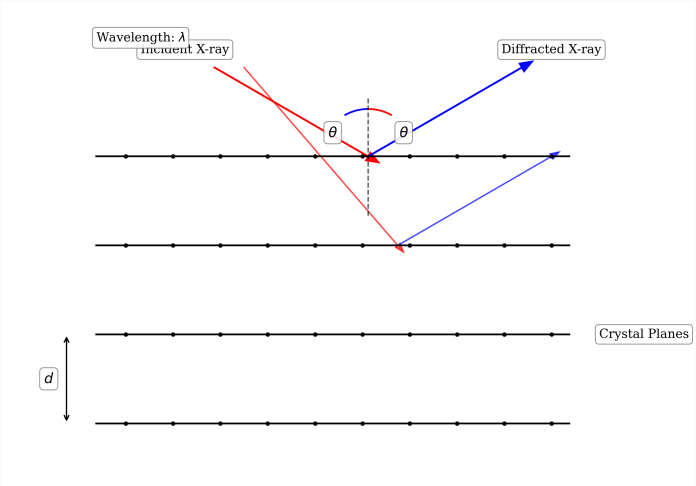

Fig. 1. Diagram for Question 1

Question 2 10.0 points

Examine the diagram below of the reciprocal lattice and Ewald sphere construction. Determine where the reflections will occur when an X-ray beam is incident on a crystal. Define the conditions under which a specific point in the reciprocal lattice satisfies the Laue condition, and interpret its meaning in the context of X-ray diffraction.

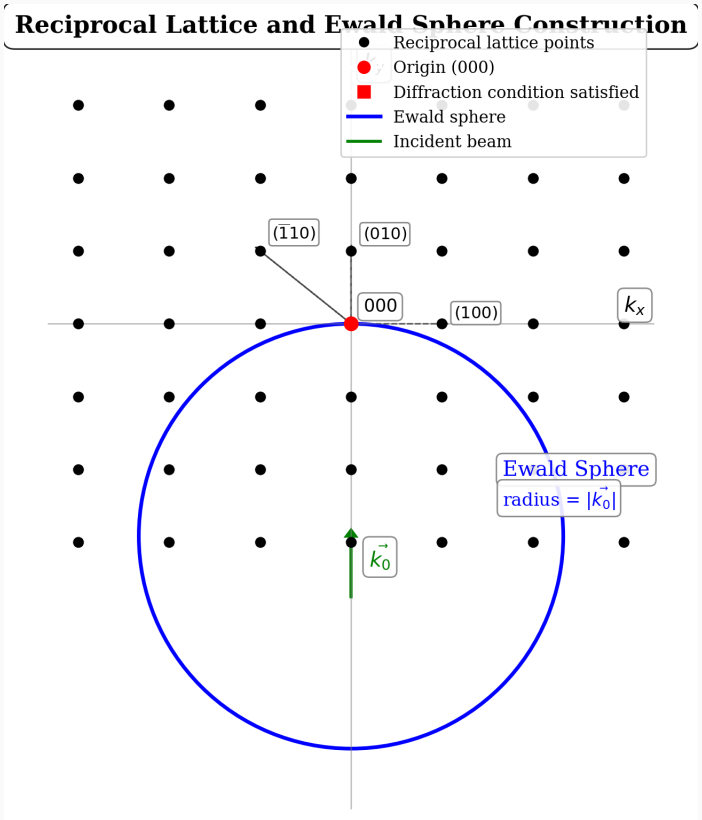

Fig. 2. Diagram for Question 2

Question 3 10.0 points

Interpret the diagram below that shows Miller indices (100), (110), and (111) planes in a cubic unit cell. Explain how these planes are oriented with respect to the cubic axes and how each set of indices determines the plane's orientation.

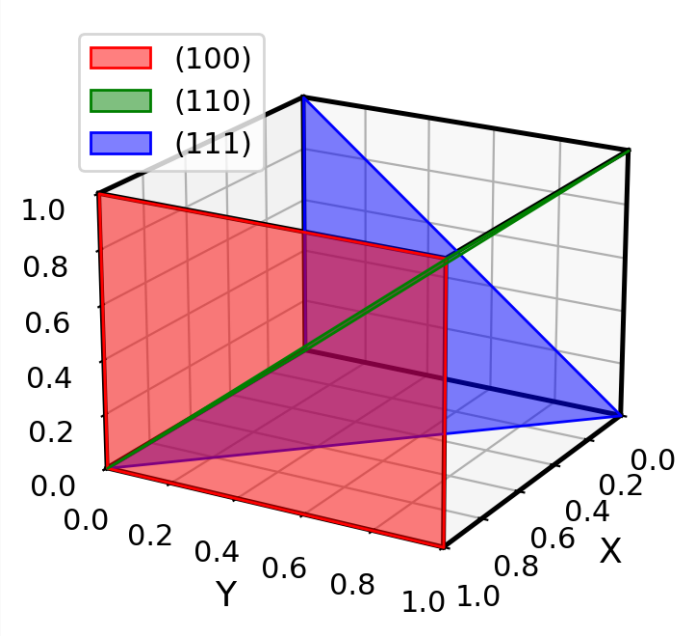

Fig. 3. Diagram for Question 3

Question 4 10.0 points

Examine the diagram below of an FCC diffraction pattern. Identify the systematic absences in the diffraction peaks and explain the underlying cause of these absences based on the lattice plane reflections.

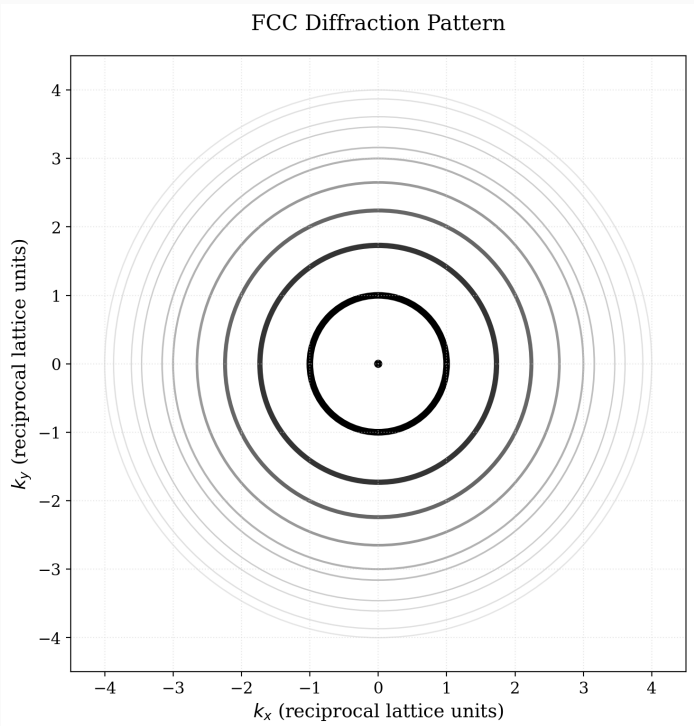

Fig. 4. Diagram for Question 4

Question 5 10.0 points

Examine the atomic arrangement of the BCC crystal in the diagram below. Use this to calculate the structure factor for a given set of Miller indices, assuming only atomic scattering factors are considered. Detail the steps involved in the calculation.

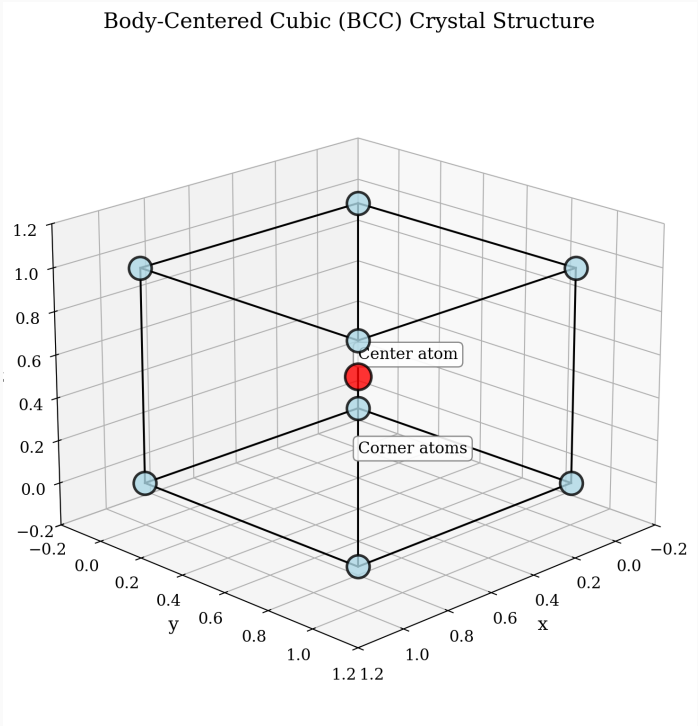

Fig. 5. Diagram for Question 5